

**Question 1:**

**Prove that  $P(A \setminus B) = P(A) - P(A \cap B)$ . Discuss what would be the result if  $B \subseteq A$ .**

**Proof:**

We are given the classical probability formula:

$$P(E) = \frac{\text{Total number of outcomes in the sample space}}{\text{Number of favorable outcomes}}$$

We also know from set theory that:

$$A = (A \cap B) \cup (A \setminus B)$$

where  $A \cap B$  and  $A \setminus B$  are disjoint sets. This means:

$$P(A) = P(A \cap B) + P(A \setminus B)$$

Rearranging this equation, we get:

$$P(A \setminus B) = P(A) - P(A \cap B)$$

**Discussion if  $B \subseteq A$ :**

If  $B \subseteq A$ , then every element of  $B$  is also in  $A$ . This implies:

$$A \cap B = B$$

Substituting this into the formula for  $P(A \setminus B)$ , we get:

$$P(A \setminus B) = P(A) - P(B)$$

Thus, when  $B \subseteq A$ , the probability of  $A \setminus B$  simplifies to the difference between  $P(A)$  and  $P(B)$ .

**Question 2:**

**Prove that if  $A$  and  $B$  are independent, then their complements  $A^c$  and  $B^c$  are also independent, i.e.,  $P(A^c \cap B^c) = P(A^c)P(B^c)$ .**

**Proof:**

We are given that  $A$  and  $B$  are independent, which means:

$$P(A \cap B) = P(A)P(B)$$

From set theory, we know:

$$A^c \cap B^c = (A \cup B)^c$$

Using De Morgan's laws, the complement of  $A \cup B$  is:

$$P((A \cup B)^c) = 1 - P(A \cup B)$$

The probability of  $A \cup B$  is given by:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Substituting  $P(A \cap B) = P(A)P(B)$  (since  $A$  and  $B$  are independent), we get:

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

Now, substituting back into  $P(A^c \cap B^c)$ :

$$P(A^c \cap B^c) = 1 - [P(A) + P(B) - P(A)P(B)]$$

Simplify:

$$P(A^c \cap B^c) = 1 - P(A) - P(B) + P(A)P(B)$$

On the other hand, the product of  $P(A^c)$  and  $P(B^c)$  is:

$$P(A^c) = 1 - P(A), P(B^c) = 1 - P(B) \quad P(A^c)P(B^c) = (1 - P(A))(1 - P(B)) = 1 - P(A) - P(B) + P(A)P(B)$$

Thus:

$$P(A^c \cap B^c) = P(A^c)P(B^c)$$

This proves that  $A^c$  and  $B^c$  are independent.

### Question 3:

**Prove that if  $A \subseteq B$ , then  $P(A \cap B) = P(A)$ .**

**Proof:**

If  $A \subseteq B$ , then every element of  $A$  is also an element of  $B$ . By definition of intersection:

$$A \cap B = A$$

Taking probabilities on both sides:

$$P(A \cap B) = P(A)$$

This completes the proof.

### Question 4:

**For three events  $A$ ,  $B$ , and  $C$ , prove that:**

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

**Proof:**

We use the principle of inclusion-exclusion for three sets. The union of three sets can be expressed as:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

This formula accounts for overcounting intersections between pairs of sets and ensures that the triple intersection is added back exactly once.

Thus, the formula is proven.

**Question 5:**

**Prove that:**

$$P(A^c|B) = 1 - P(A|B)$$

**Proof:**

From the definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Using the fact that  $A^c \cup A = S$  (the entire sample space), we have:

$$P(A^c \cap B) + P(A \cap B) = P(B)$$

Rearranging:

$$P(A^c \cap B) = P(B) - P(A \cap B)$$

Substitute this into the formula for  $P(A^c|B)$ :

$$P(A^c|B) = \frac{P(B) - P(A \cap B)}{P(B)}$$

Split the fraction:

$$P(A^c|B) = 1 - \frac{P(A \cap B)}{P(B)}$$

From the definition of conditional probability,  $\frac{P(A \cap B)}{P(B)} = P(A|B)$ . Thus:

$$P(A^c|B) = 1 - P(A|B)$$

This completes the proof.

**Final Answer:**

All proofs are complete. If you need further clarification on any step, feel free to ask!